

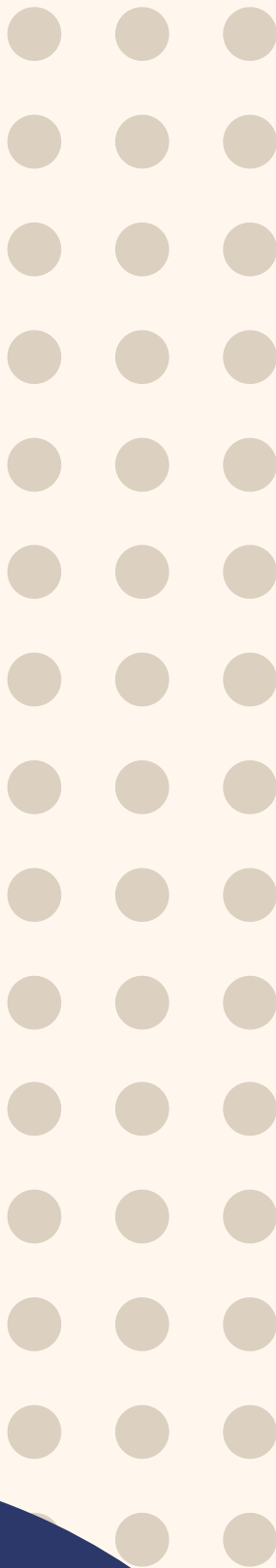
RIYADH RESTAURANTS



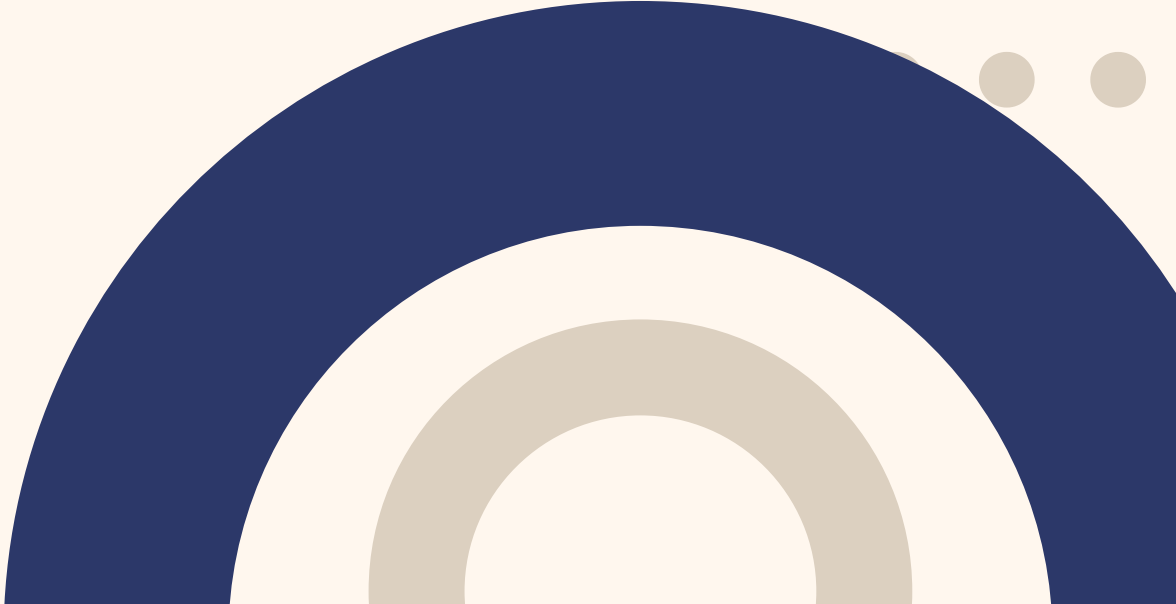
Presented Group 5

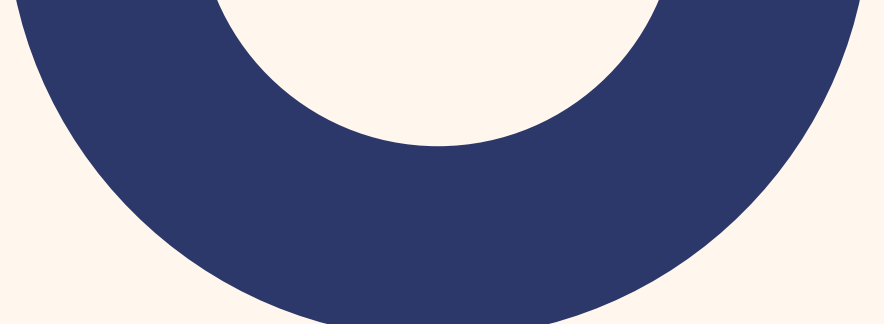
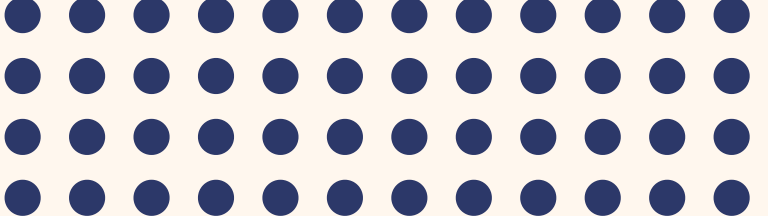


Table of Contents

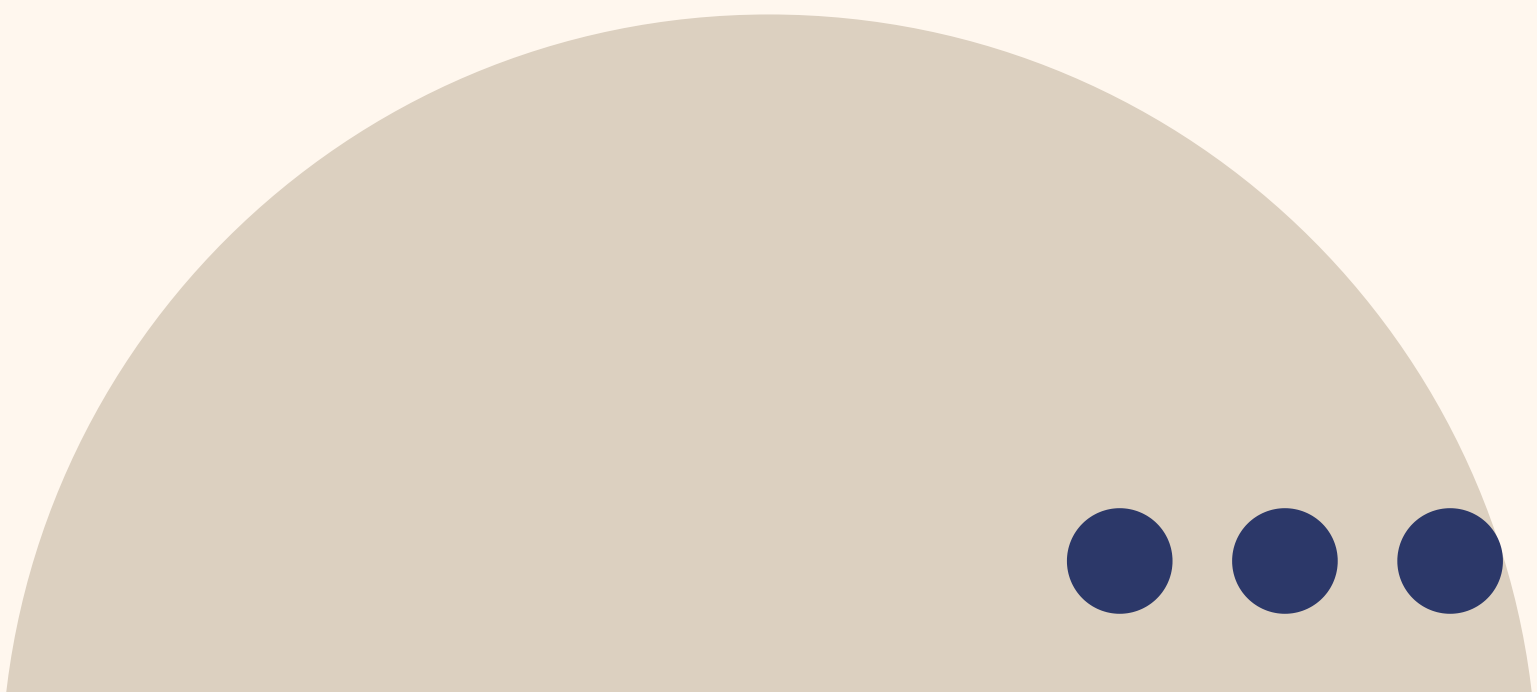


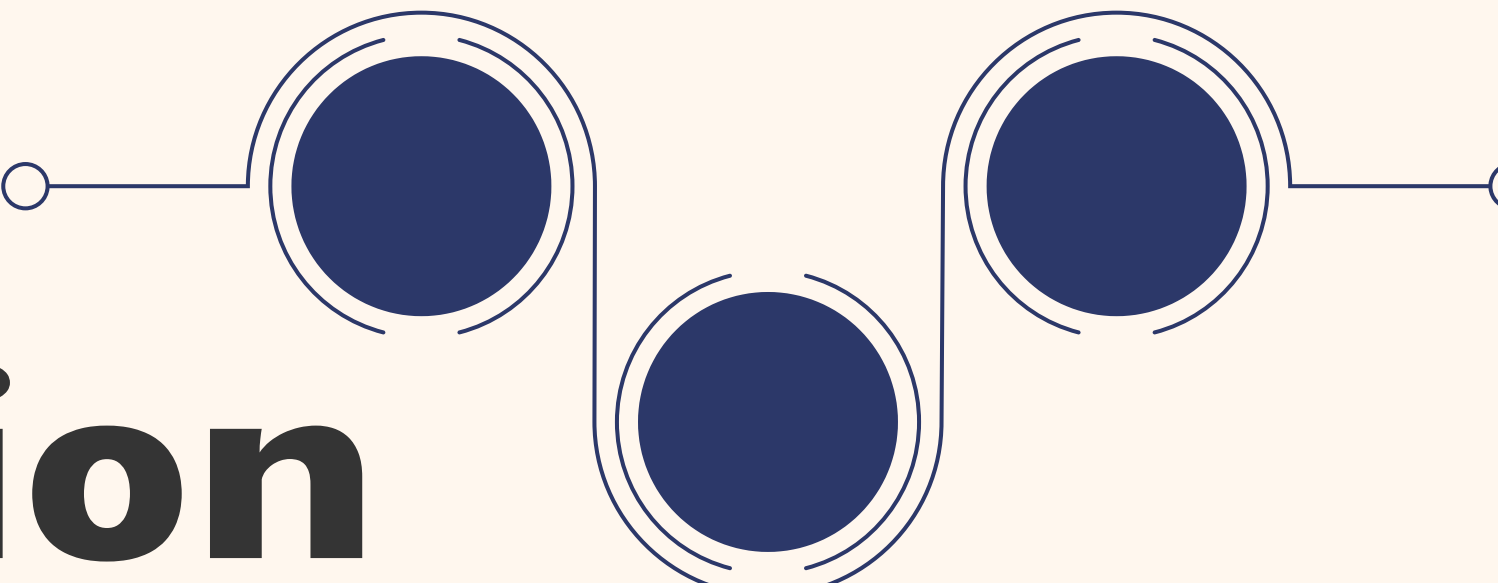
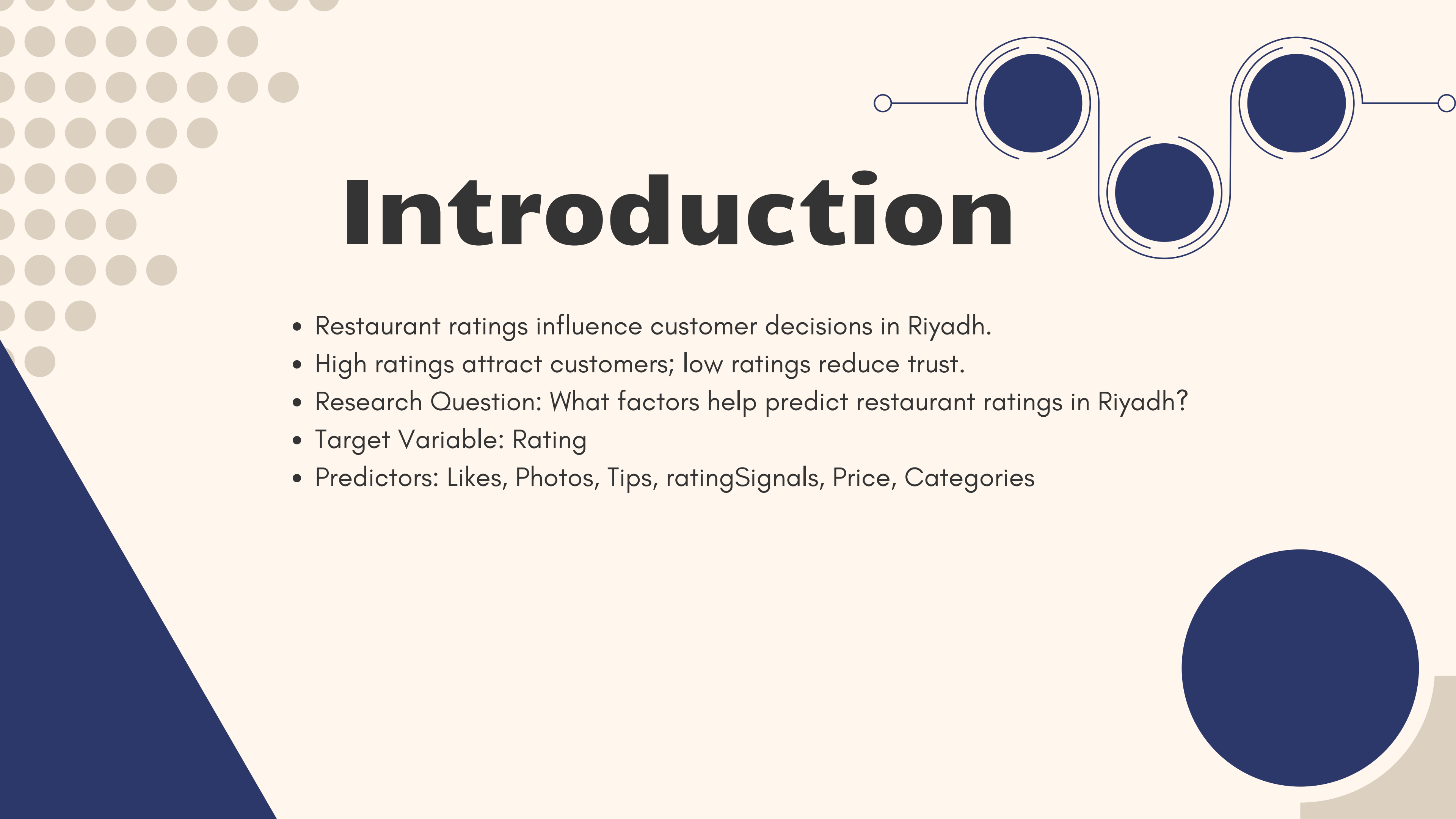
- *Abstract*
- **Introduction**
- **Dataset**
- **Data Preprocessing**
- **Exploratory Data Analysis**
- **Modeling Approach**

- *Methodology*
 - **Results and Evaluation**
 - **Insights and Recommendations**
 - *Reference*
- 



Abstract

- Analyzed the Riyadh Restaurants 20K dataset.
 - Research question: Which restaurant features help predict customer rating?
 - Target variable: Rating
 - Predictors: Likes, Photos, Tips, ratingSignals
 - Applied data cleaning, exploratory analysis, and linear regression.
 - Engagement features were stronger predictors than descriptive information
- 
- 



Introduction

- Restaurant ratings influence customer decisions in Riyadh.
- High ratings attract customers; low ratings reduce trust.
- Research Question: What factors help predict restaurant ratings in Riyadh?
- Target Variable: Rating
- Predictors: Likes, Photos, Tips, ratingSignals, Price, Categories

Dataset

- Dataset: Riyadh Restaurants 20K
- Includes restaurant information such as name, category, location, likes, photos, tips, rating, and ratingSignals.
- Each row represents one restaurant.
- Target variable ranges from 0 to 5.
- Dataset contains about 20,000 rows and 10–15 columns.



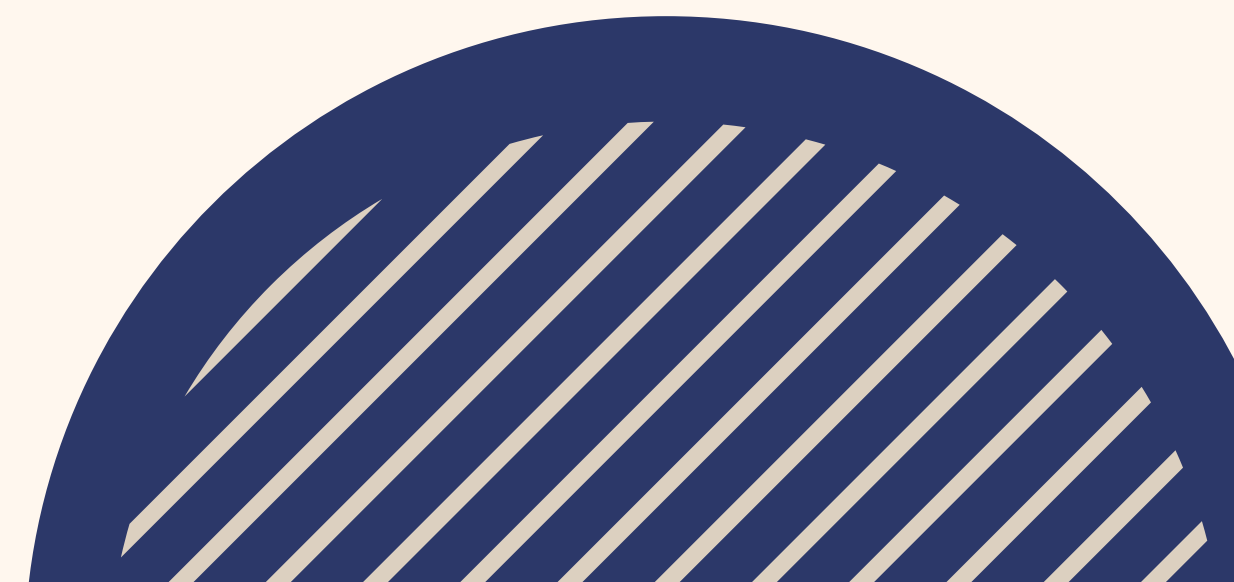
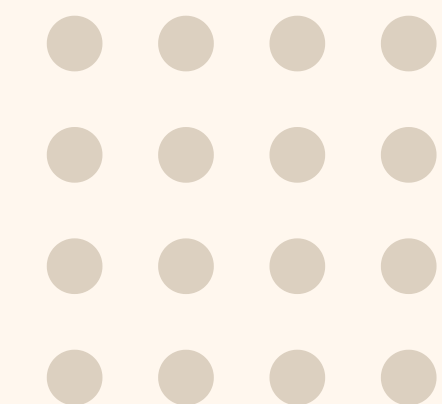
Variables

variables	why included
Rating	Target variable
Likes	Customer engagement
Photos	Online visibility
Tips	Customer interaction
ratingSignals	Rating reliability
Price and Categories	Restaurant characteristics
Lat/Lng	Geographic location



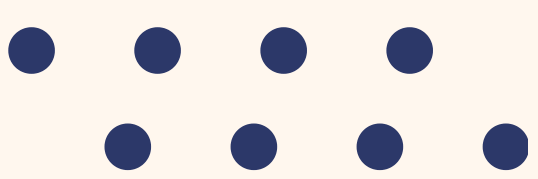
Sampling and Filtering

- Removed rows with missing rating values.
- Selected numeric predictors: likes, photos, tips, ratingSignals.
- Categories and price used mainly for visualization.
- Final dataset size: 19,361 rows.
- Training rows: 15,488
- Test rows: 3,873



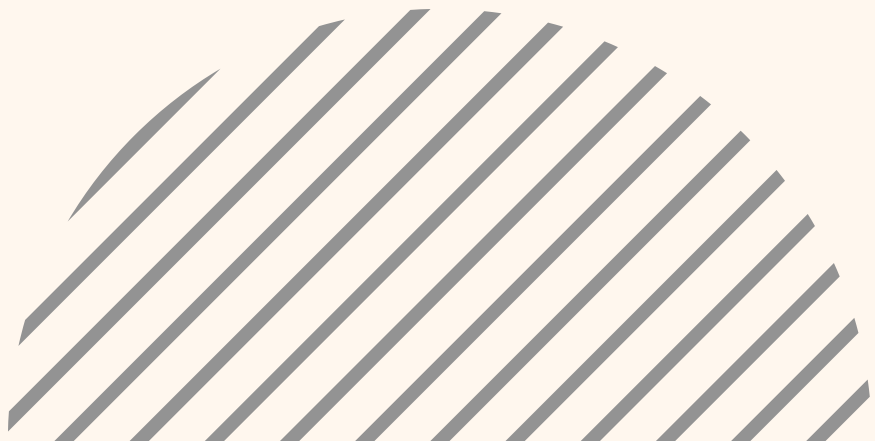
Data Preprocessing

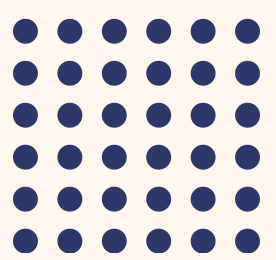
- Handled missing values in rating and predictor columns.
- Converted numeric columns into correct format.
- Categorical variables were excluded from first regression model.
- No major new features were created.



Exploratory Data Analysis

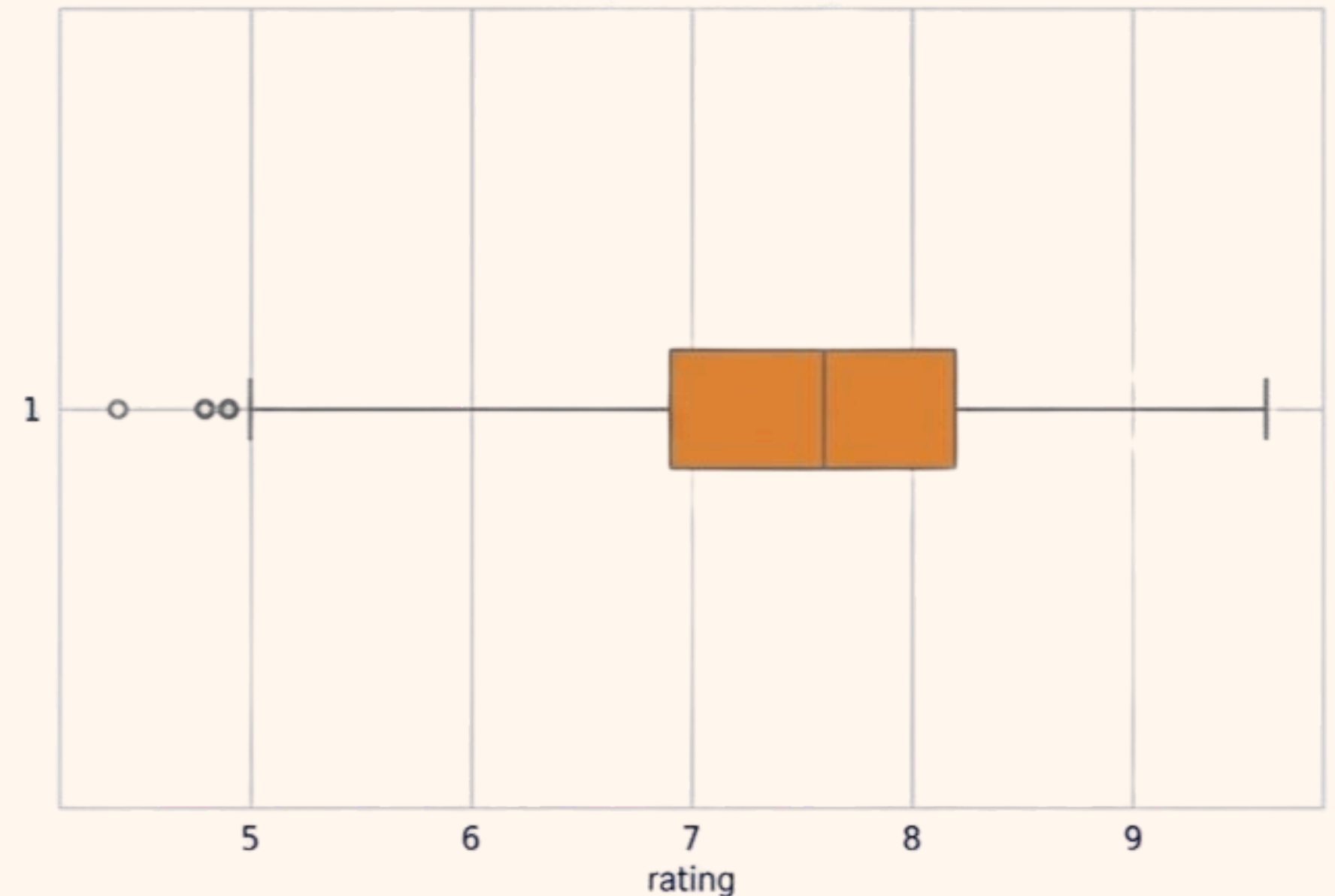
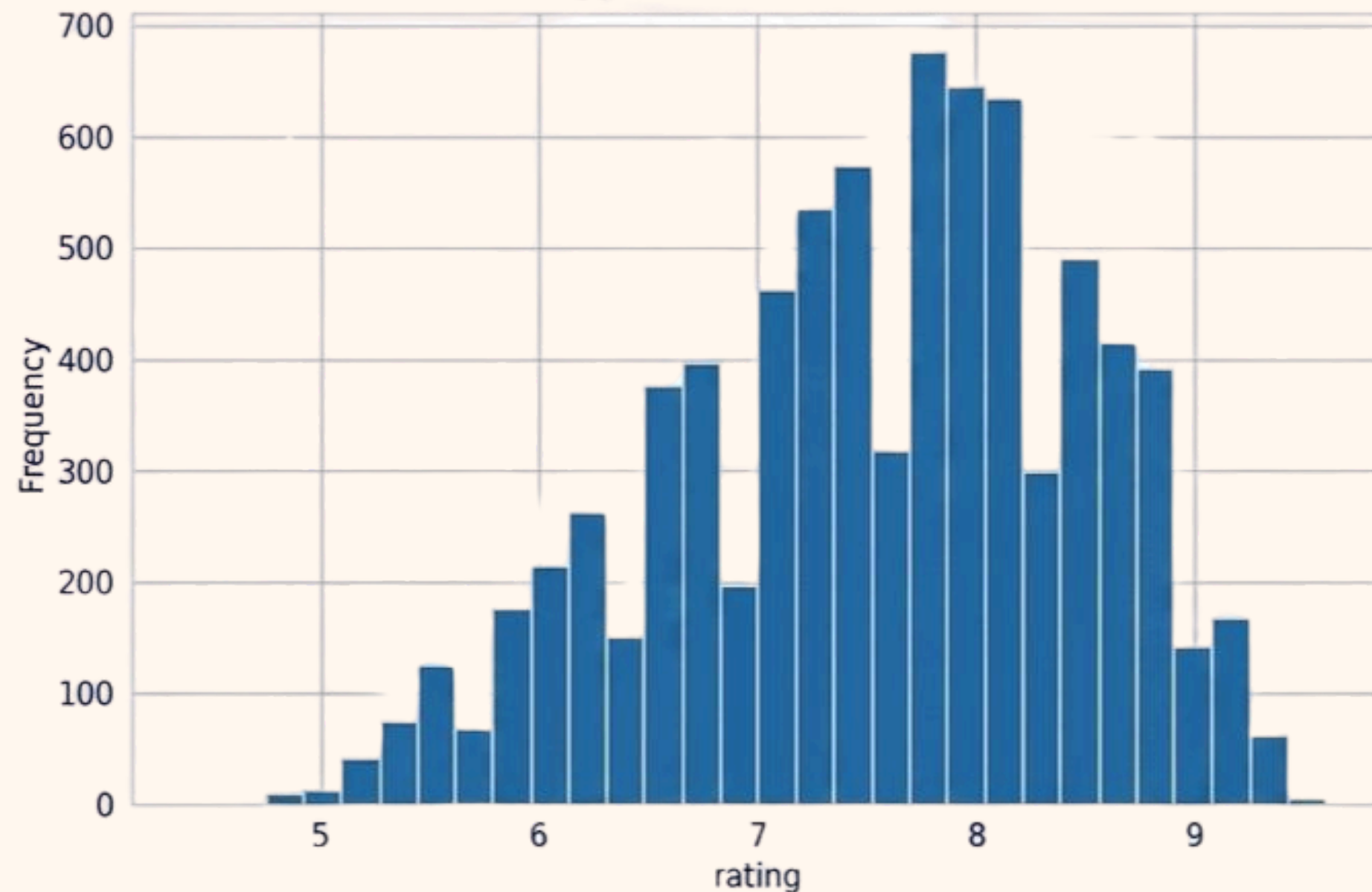
- Analyzed rating distribution.
- Compared ratings across price levels and categories.
- Examined relation
- nships with likes, photos, tips, and ratingSignals.





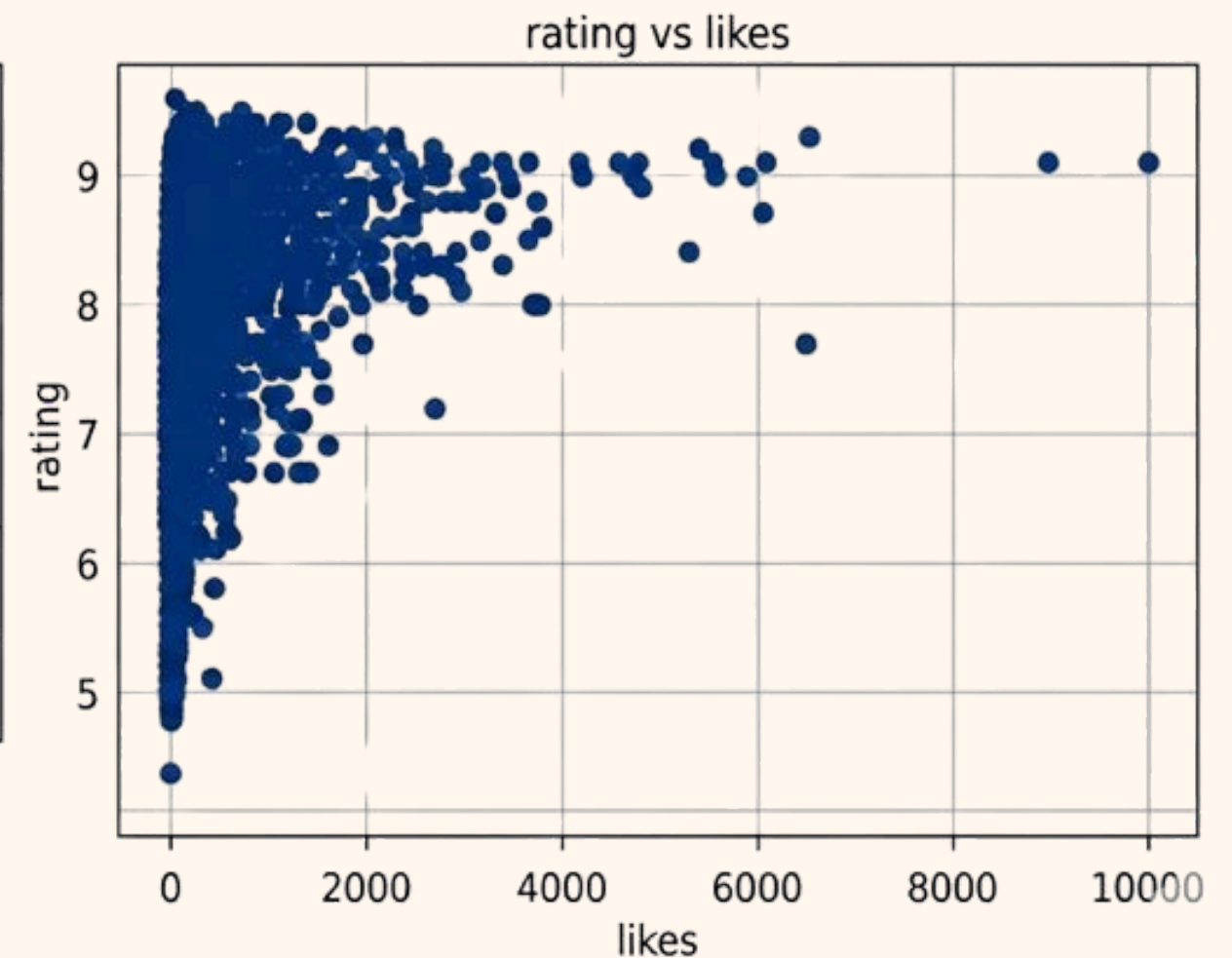
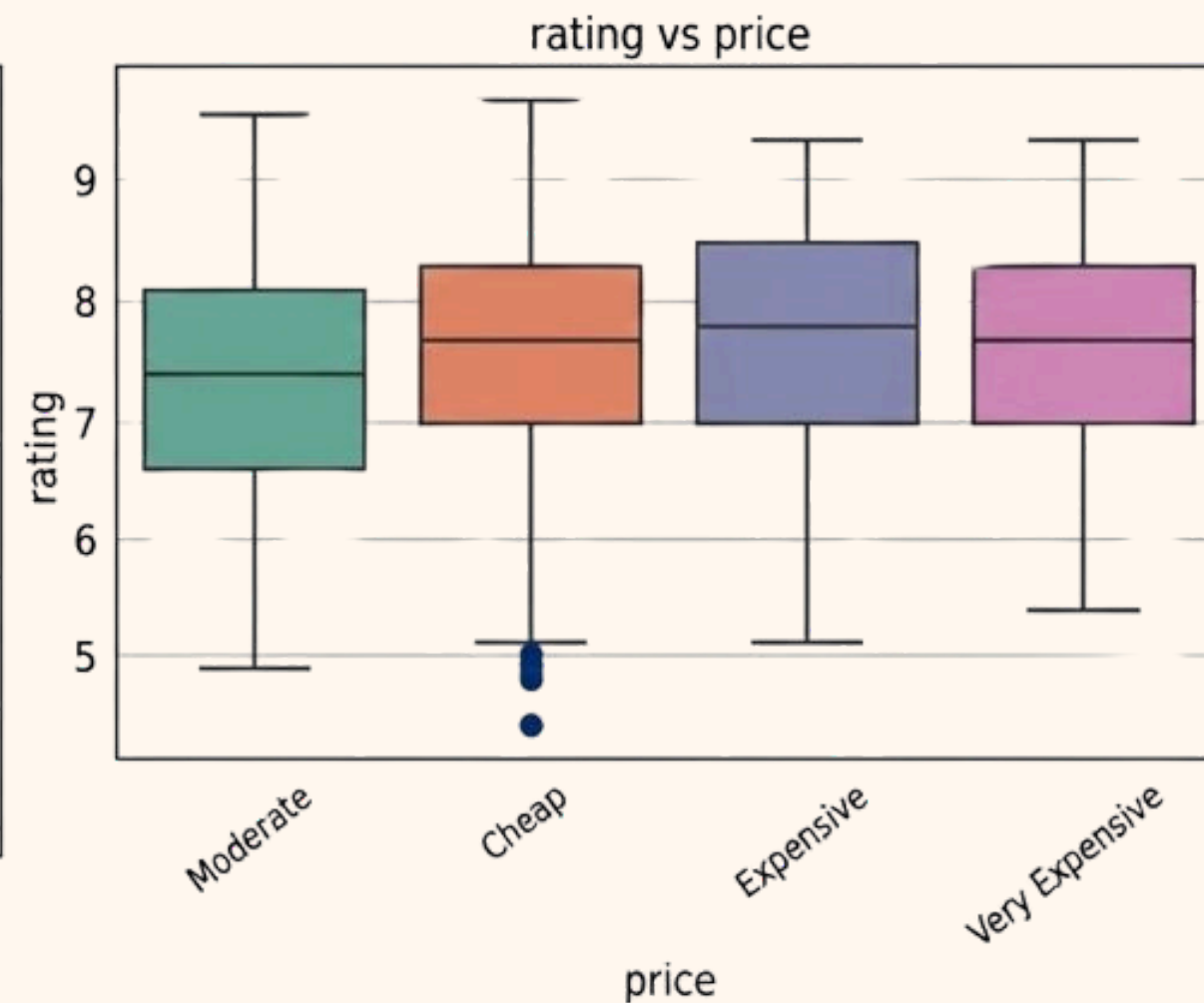
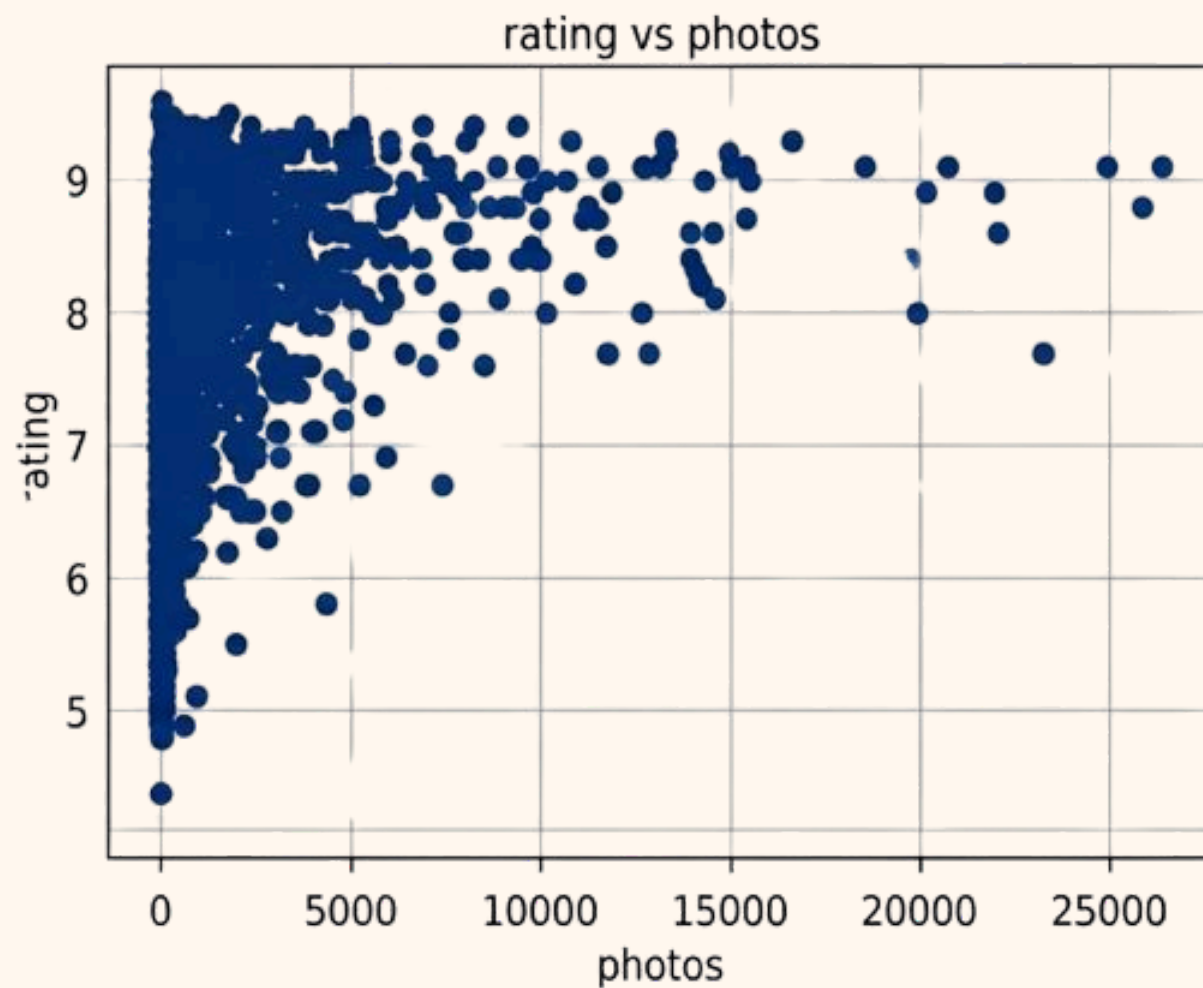
Distribution of Ratings

- Ratings concentrated between 7 and 9.
- Average rating approximately 7.54.
- Boxplot showed a few low-rating outliers.
- Most restaurants had higher ratings.



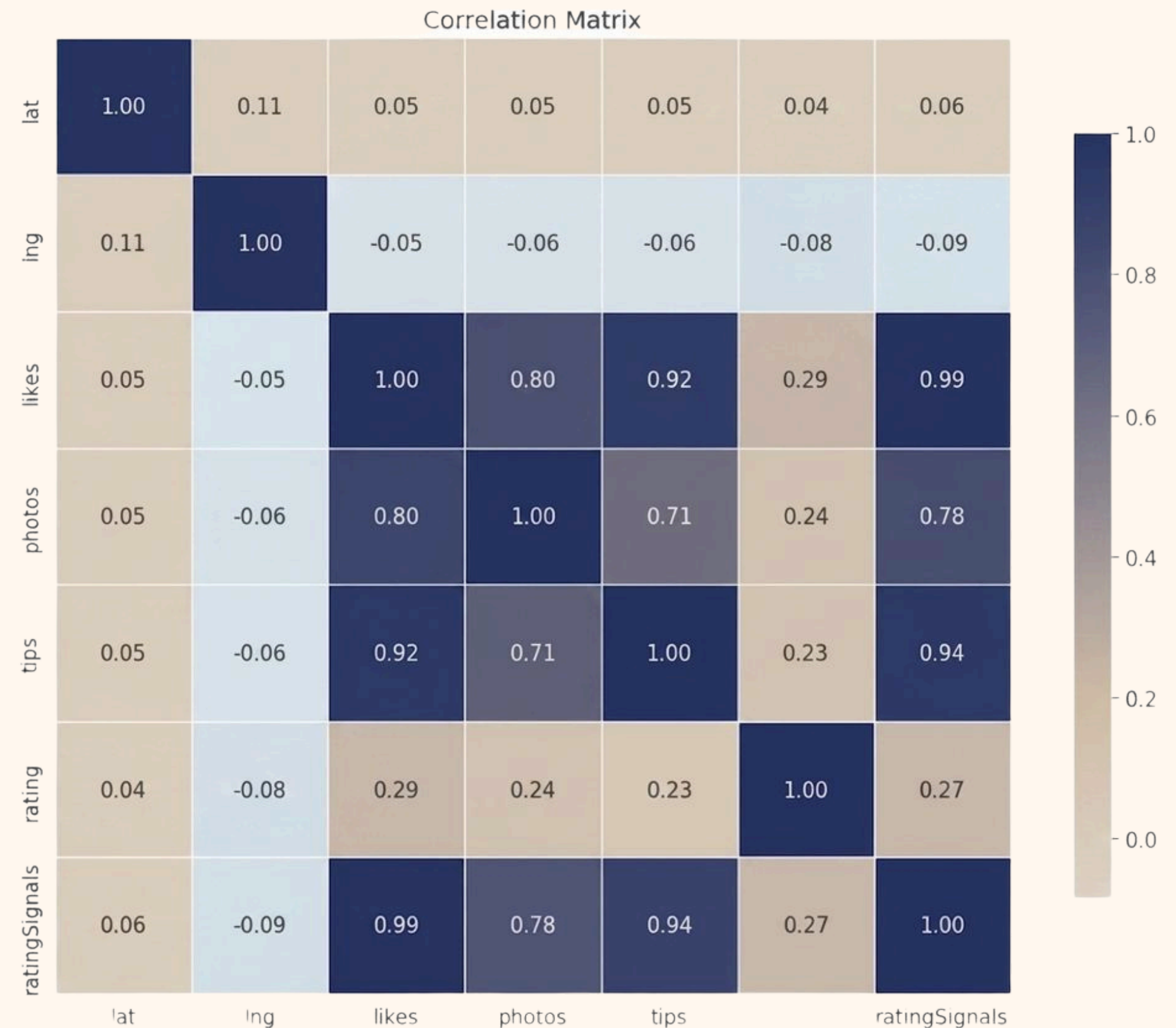
Key Relationships

- Relationships explored between rating and price, likes, and photos.
- Restaurants with higher online activity tended to have higher ratings.
- Scatter plots and boxplots were used.



Correlation Analysis

- Heatmap examined relationships between numeric variables.
- Identified strongest predictors related to rating.
- Checked for possible collinearity.

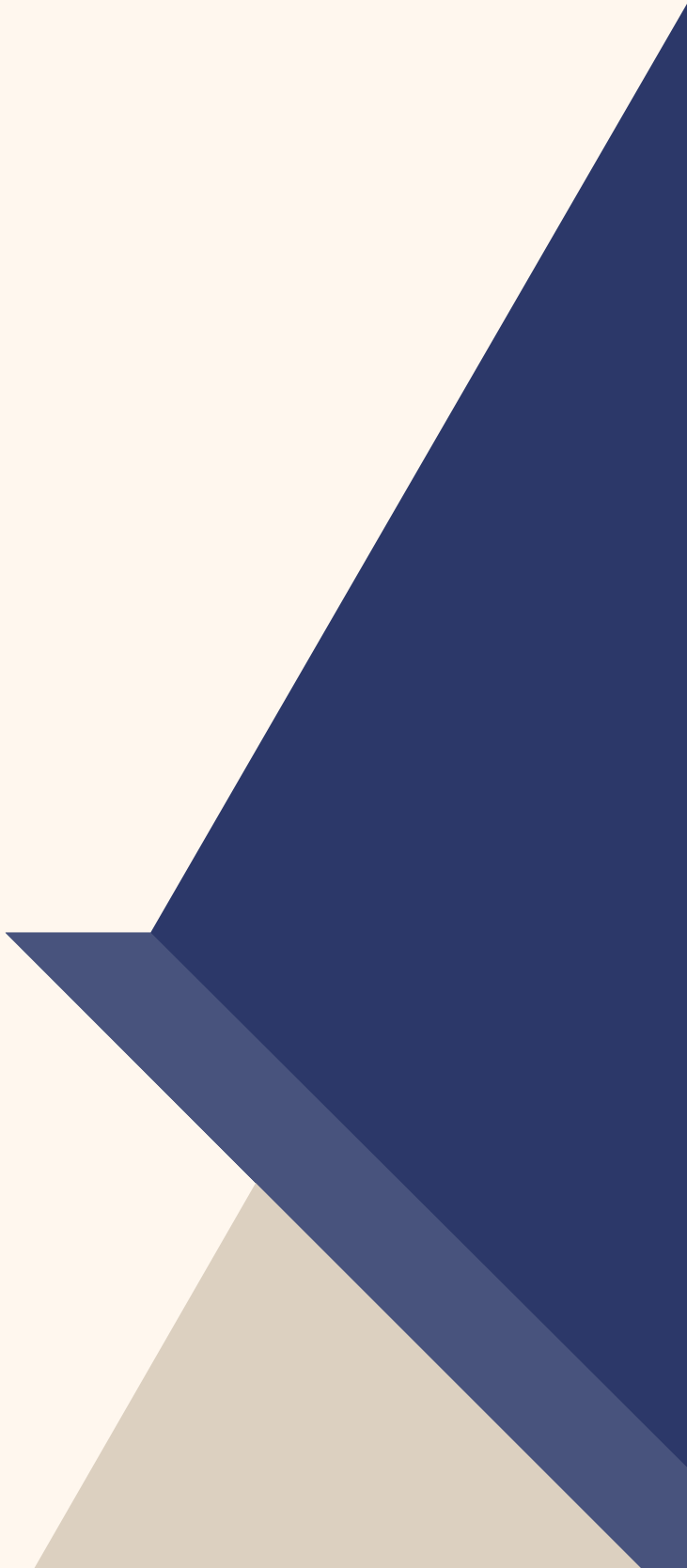


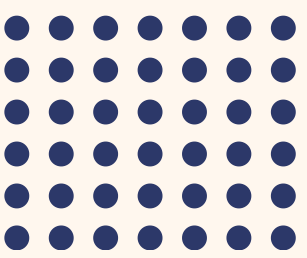


Modeling Approach

- Used Multiple Linear Regression.
- Predictors: likes, photos, tips, ratingSignals.
- Categories and price excluded due to encoding requirements.

Train/ Test Split

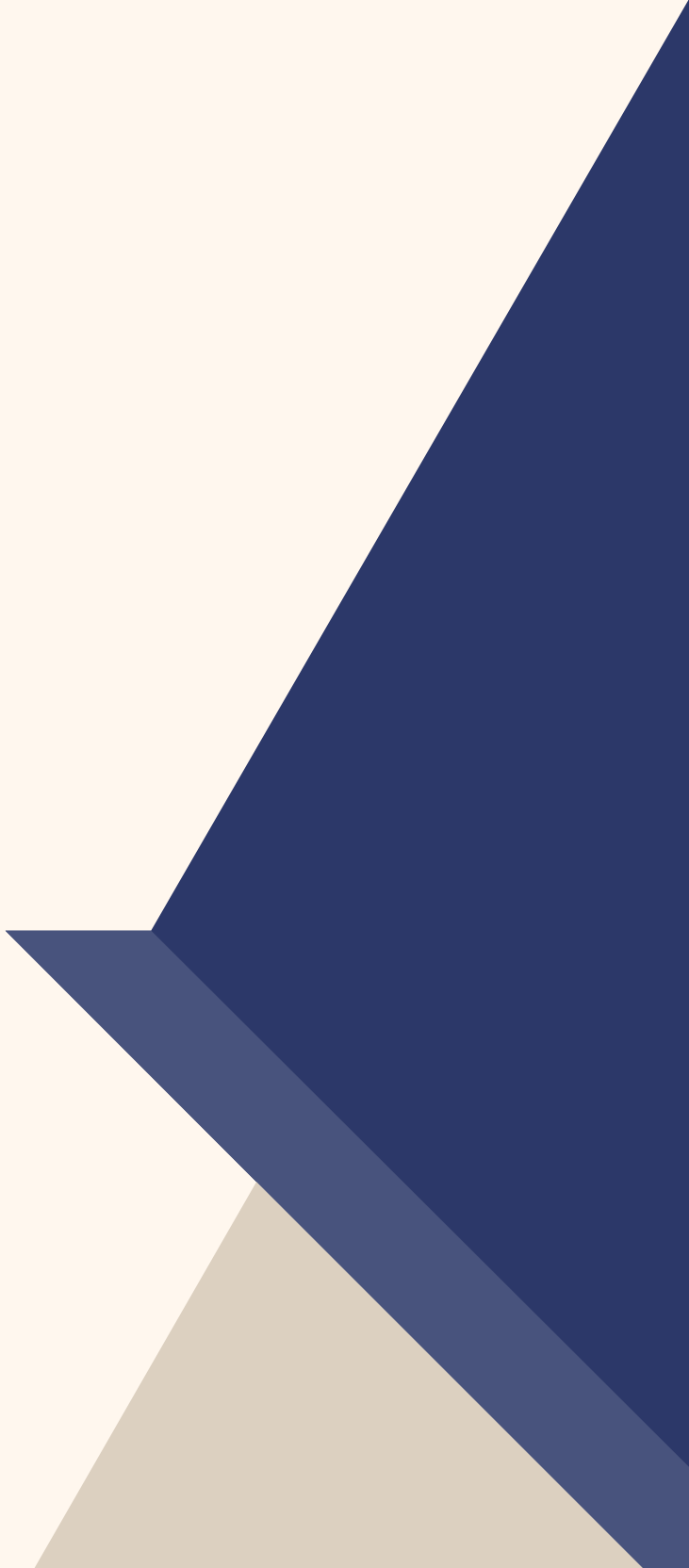
- Used 80/20 split.
 - Training set: 15,488 rows.
 - Test set: 3,873 rows.
 - No stratification used.
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Results and Evaluation

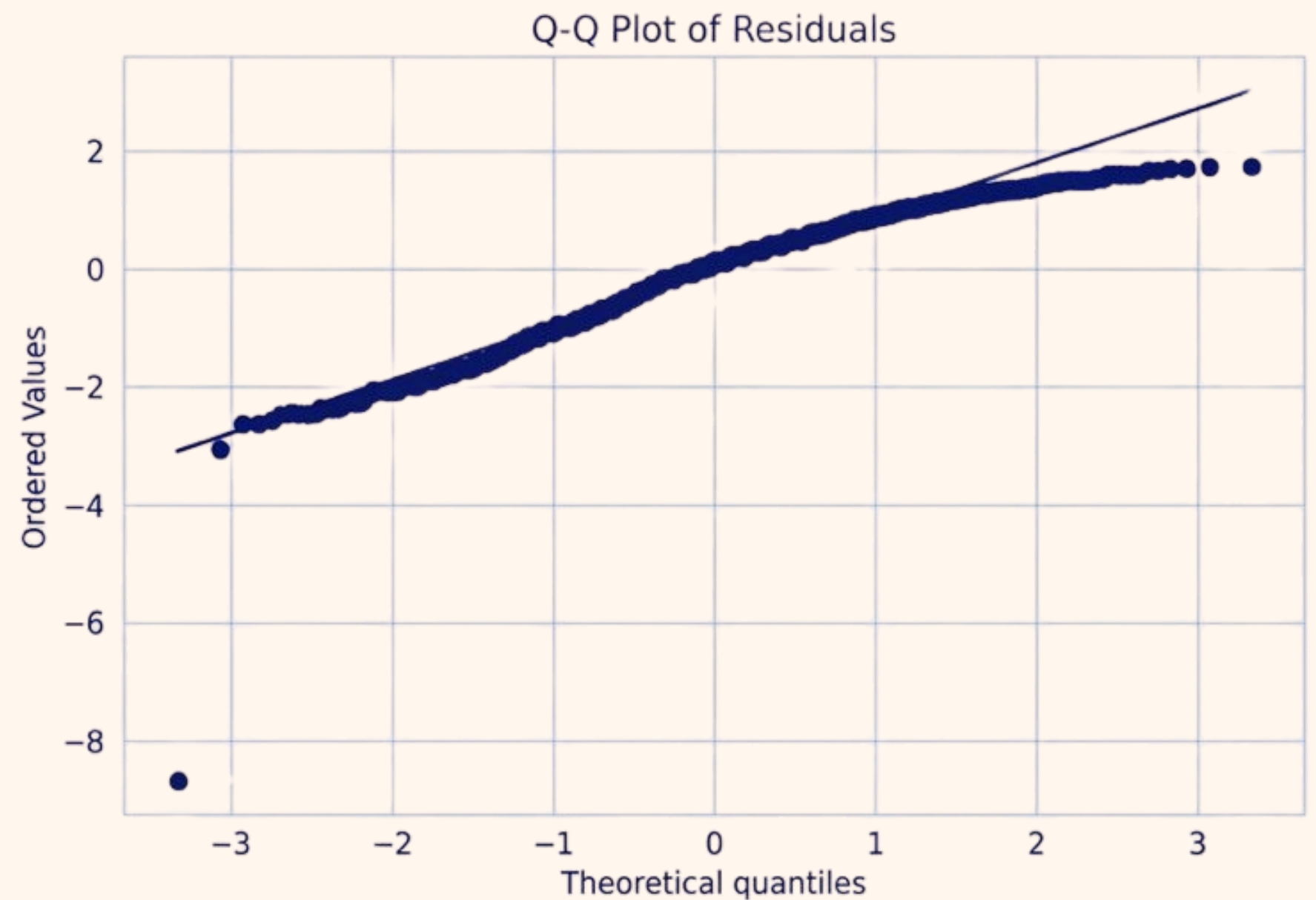
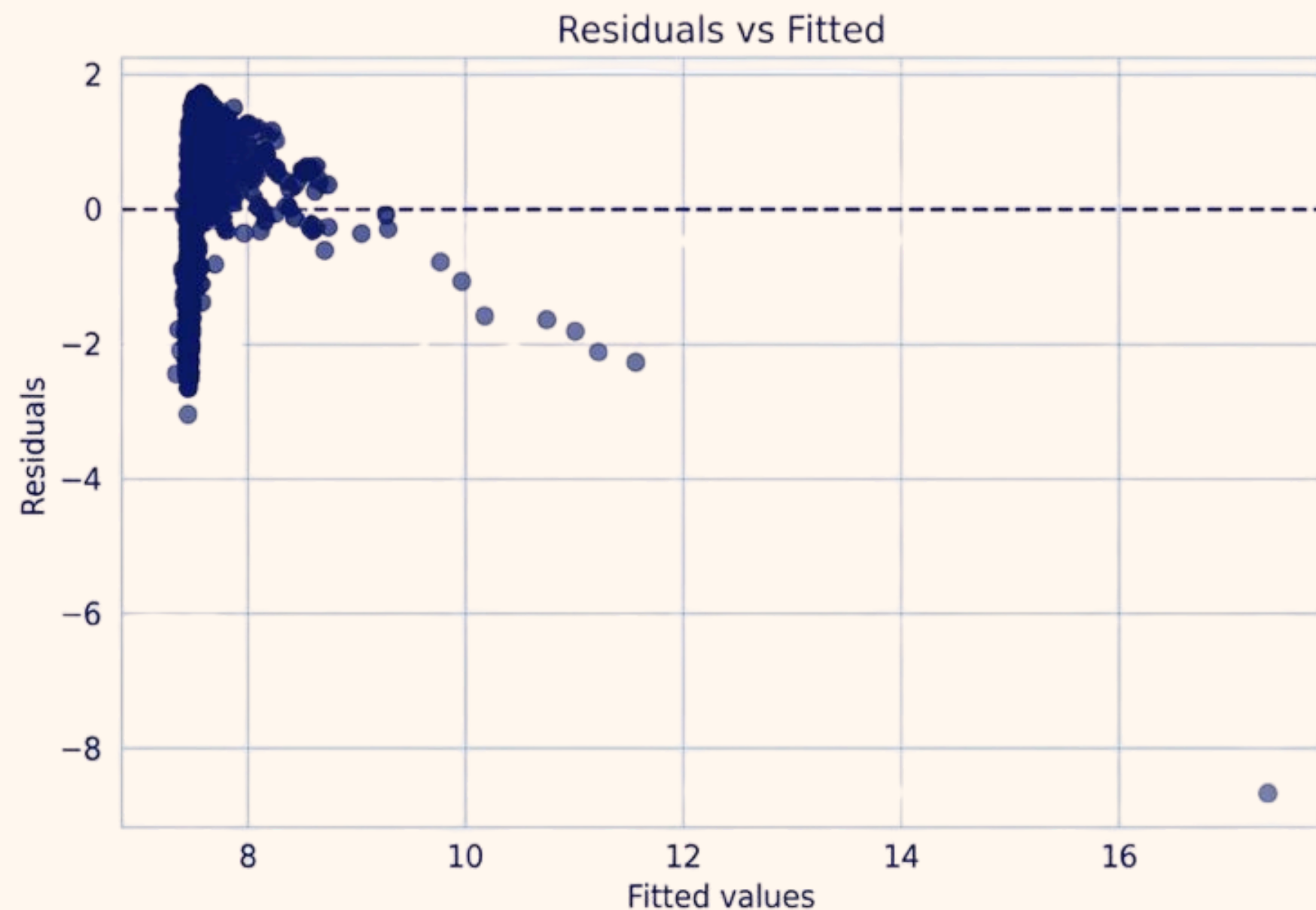
- Evaluation metrics: R-squared, RMSE, MAE.
- Test R-squared: 0.0679
- Test RMSE: 0.9378
- Test MAE: 0.7453
- Model provides a simple baseline.

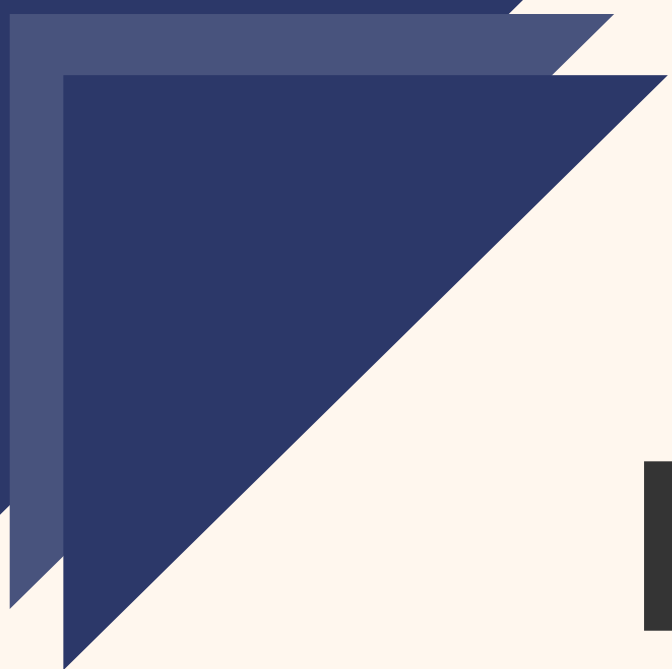
Performance Metrics

- Training R-squared: 0.0962
 - Test R-squared: 0.0679
 - Training RMSE: 0.8893
 - Test RMSE: 0.9378
 - Training MAE: 0.7209
 - Test MAE: 0.7453
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Diagnostic Checks

- Residuals were not perfectly random.
- Q-Q plot showed deviations from normality.
- Model useful as a baseline with limited predictive strength.





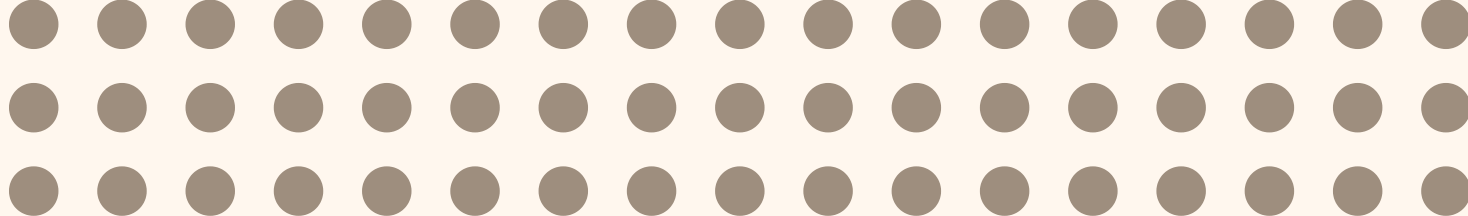
Insights and Recommendations

- Restaurants with more likes and photos tend to have higher ratings.
- Restaurant owners should improve online presence.
- Customer engagement may support better ratings.

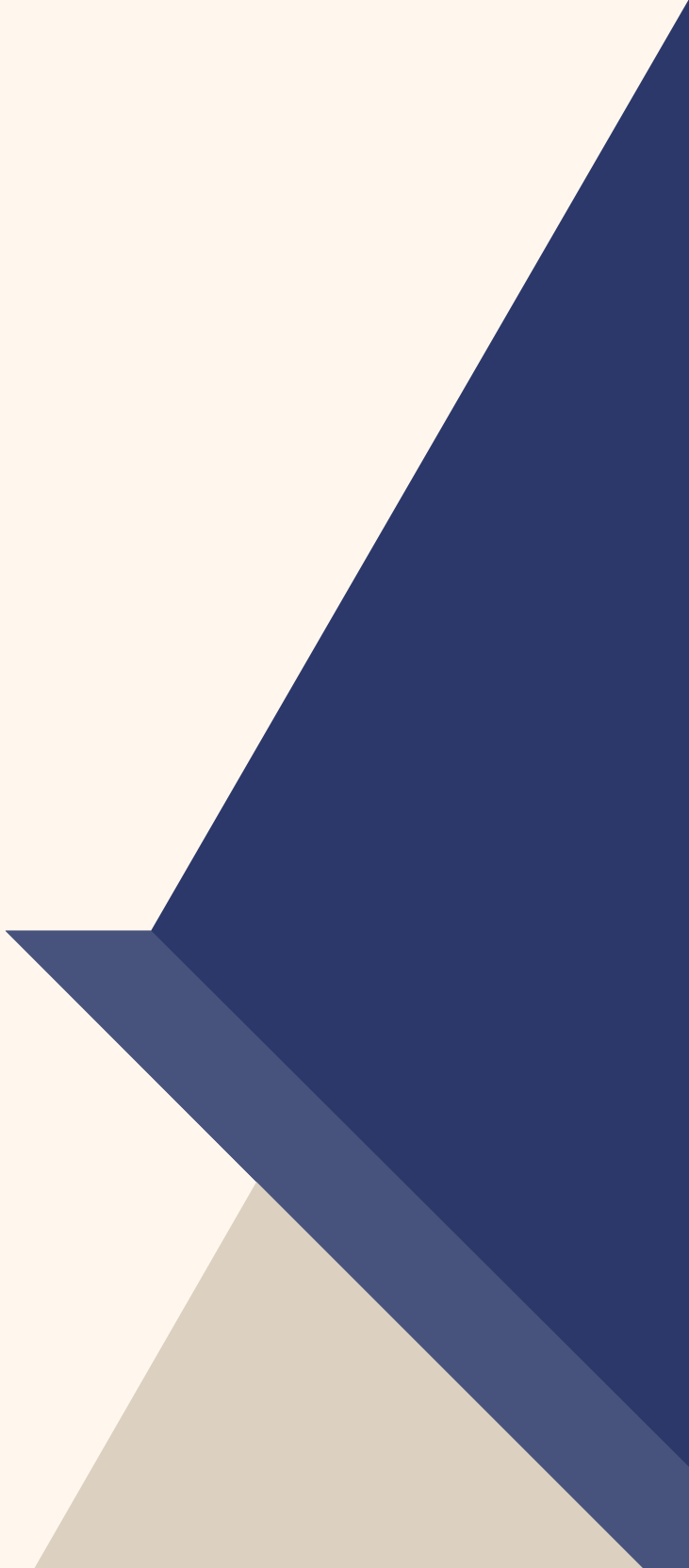
Limitations and Future Work

- Model used only numeric variables.
- Future work may include encoded categorical variables.
- Review text analysis may improve prediction accuracy.





Reference

- Riyadh Restaurants 20K Dataset — Kaggle
 - scikit-learn: Machine Learning in Python
 - CS474 Course Materials
 - Dr.Hamad Al-Abduljabbar
- 



THANK YOU

For your attention

